BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Insertion Sort

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. When does ascending bubble sort swap two adjacent items?

2. What does readLine() do on an open text file?

A) Writes a new line
B) Deletes the next line
C) Reads the next line
D) Closes the file

3. "Primary storage exists only to keep long-term backups." Correct this. Explain why.

4. "My unsaved work will survive a power cut because this computer has ROM." Explain the mistake. How could the work have been kept?

Today's objectives

Explain the steps and required conditions for insertion sort. Apply the algorithm or method to supplied data.

Key vocabulary - Read, cover, recall

Insertion sort
Builds a growing sorted prefix by inserting each next value.

Key
Current element being inserted.

Sorted prefix
Left section already in order.

Shift
Move an existing value one place right to make room.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Build a sorted prefix	Treat the first element as a sorted section, then take the next element as a key.
Make space	Move larger sorted elements one position right to create room for the key.
Insert the key	Place the key in the correct position in the existing sorted prefix.
Grow the prefix	Repeat with each remaining element until the whole list is ordered.

I Do - One worked example

Trace insertion sort on [7,3,5,1,4], showing the sorted prefix after each insertion. [5 marks]

1. Insert 3

2. Move 7 right; result [3,7,5,1,4].

3. Insert 5 -> [3,5,7,1,4]; insert 1 -> [1,3,5,7,4].

4. Insert 4 -> [1,3,4,5,7]. All five values are sorted. [5]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace the first two insertions of [6,2,4,1]. [4]

2. Insert key 3 into sorted prefix [1,4,6] (next value 3). [3]

3. Explain why insertion sort can be efficient on nearly sorted data. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the line that shifts the larger value during insertion.

```
array depths = [16,8]
key = depths[1]
if depths[0] > key then
...
depths[0] = key
endif
print(depths[1])
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. After inserting the first two unsorted items from [5,2,4,3], state the array at each stage. [4]

Q3. Explain the purpose of the sorted section in insertion sort. [3]

Q4. A sorted prefix is [2,6,8] and next key is 5. State the resulting prefix after insertion. [2]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Binary search can safely discard half only when data is sorted. ☐ T ☐ F
2. Treat the first element as a sorted section, then take the next element as a key. ☐ T ☐ F
3. Move larger sorted elements one position right to create room for the key. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 When the left item is greater than the right item. ☐

Q2 C - Reads the next line ☐

Q3 It holds instructions and working data that the CPU needs while programs run. ☐

Q4 Unsaved work is in volatile RAM, so it is lost when power goes off. Save it to non-volatile secondary storage to keep it. ☐

Code First - Model

```
array depths = [16,8]
key = depths[1]
if depths[0] > key then
depths[1] = depths[0]
depths[0] = key
endif
print(depths[1])
```

Output: 16

We Do - Check the model

1. After inserting 2: [2,6,4,1] [2]; after inserting 4: [2,4,6,1] [2]. ☐

2. Shift 6 right [1], shift 4 right [1]; insert 3 after 1 [1], prefix [1,3,4,6]. ☐

3. Many keys are near their correct position [1], so fewer shifts/comparisons are needed [1] than when elements are far out of order [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Insert 2 -> [2,5,4,3] [2]; insert 4 -> [2,4,5,3] [2]. ☐

Q3 Contains elements already in order [1]; next key is placed at its appropriate position [1] to grow sorted section [1]. ☐

Q4 [2,5,6,8] [2]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: